



2018 HSC Course Assessment Task 3 (Trial Examination)

Wednesday 27th of June, 2018

- Working time – 3 hours.
(plus 5 minutes reading time)
- Write using blue or black pen. Where diagrams are to be sketched, these may be done in pencil.
- Board approved calculators may be used.
- Attempt **all** questions.
- At the conclusion of the examination, bundle the booklets + answer sheet used in the correct order within this paper and hand to examination supervisors.

- Mark your answers on the answer sheet provided (numbered as page 13)

- Commence each new question on a new page. Write on both sides of the paper.
- All necessary working should be shown in every question. Marks may be deducted for illegible or incomplete working.

Class (please ✓)

○ 12M4C – Mr Ireland

[illegible]

Section I

10 marks

Attempt Question 1 to 10

Allow approximately 15 minutes for this section

Mark your answers on the answer sheet provided.

Questions	Marks
1. Let $z = \sqrt{3} + i$. What is the value of $\overline{\left(\frac{i}{z}\right)}$?	1
(A) $1 - i\sqrt{3}$	(C) $\frac{-1 + i\sqrt{3}}{4}$
(B) $\frac{1 - i\sqrt{3}}{4}$	(D) $\frac{\sqrt{3} - i}{4}$
2. An ellipse has foci at $(-5, 0)$ and $(5, 0)$ and its directrices have equations $x = -10$ and $x = 10$. What is the eccentricity of the ellipse?	1
(A) $\frac{1}{\sqrt{2}}$	(C) 2
(B) $\sqrt{2}$	(D) $\frac{1}{2}$
3. Given that $3x^3 - 5x + 6 = 0$ has roots α , β and γ . What is the value of $\alpha^3 + \beta^3 + \gamma^3$?	1
(A) -1	(C) 2
(B) 6	(D) -6
4. Which of the following, for $x > 0$, is an expression for $\int \frac{2}{x + x^3} dx$?	1
(A) $\ln x\sqrt{1 + x^2} + C$	(C) $\ln \frac{x^2}{x^2 + 1} + C$
(B) $\ln x^2(1 + x^2) + C$	(D) $\ln \frac{x^2}{\sqrt{1 + x^2}} + C$

5. Using a suitable substitution, what is the correct expression for $\int_0^{\frac{\pi}{3}} \sin^4 x \cos^3 x \, dx$ in terms of u ? 1

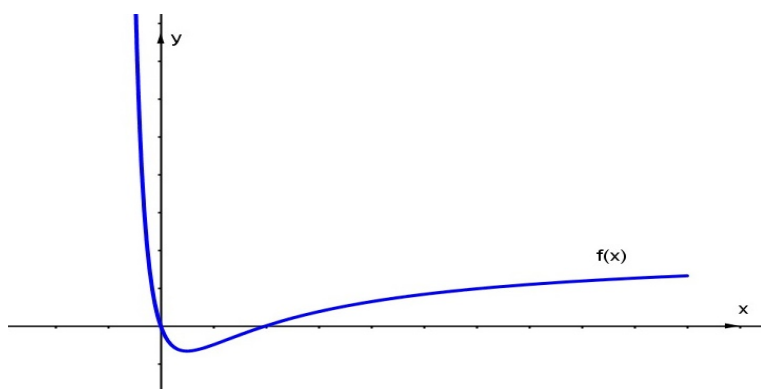
(A) $\int_0^{\frac{\sqrt{3}}{2}} (u^6 - u^4) \, du$

(C) $\int_{\frac{1}{2}}^1 (u^4 - u^6) \, du$

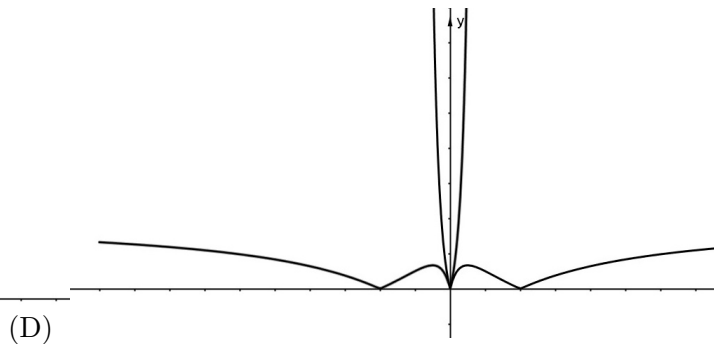
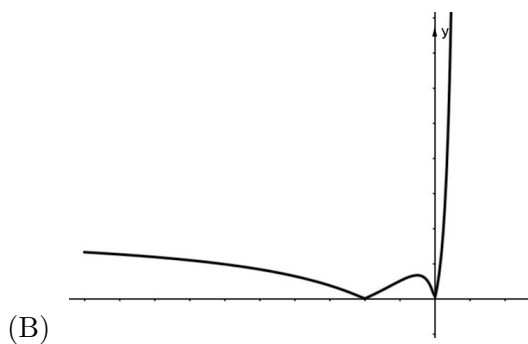
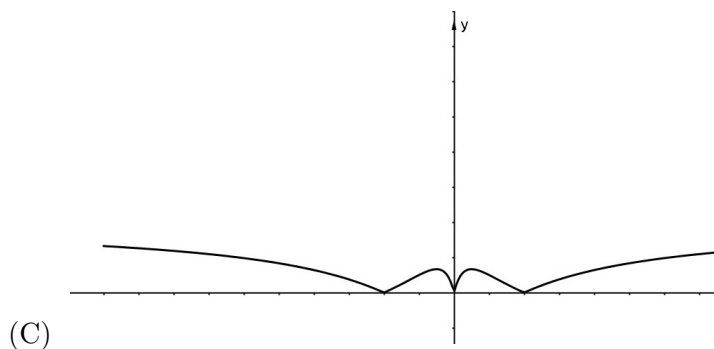
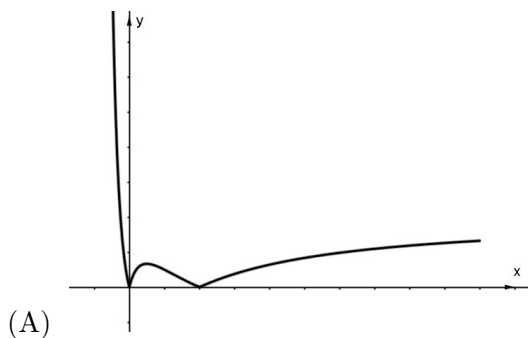
(B) $\int_1^{\frac{1}{2}} (u^4 - u^6) \, du$

(D) $\int_0^{\frac{\sqrt{3}}{2}} (u^4 - u^6) \, du$

6. Consider the graph of $y = f(x)$ drawn below. 1



Which of the following diagrams shows the graph of $|f(-x)|$?



7. Using implicit differentiation on the equation $y^3 = 3x^2y - 2x^3$, then $\frac{dy}{dx}$ would equal 1

(A) $\frac{-2x^2}{y^2 - x^2}$

(C) $\frac{2x}{x - y}$

(B) $\frac{2x}{x + y}$

(D) $\frac{y^2 - x^2}{2x^2}$

8. The normal to the point $P\left(cp, \frac{c}{p}\right)$ on the rectangular hyperbola $xy = c^2$ has the equation $p^3x - py + c - cp^4 = 0$. The normal cuts the hyperbola at another point $Q\left(cq, \frac{c}{q}\right)$. What is the relationship between p and q ? 1

(A) $pq = -1$

(C) $p^4q = -1$

(B) $p^2q = -1$

(D) $p^3q = -1$

9. ω is a non-real root of the equation $z^5 + 1 = 0$. Which of the following is not a root of this equation? 1

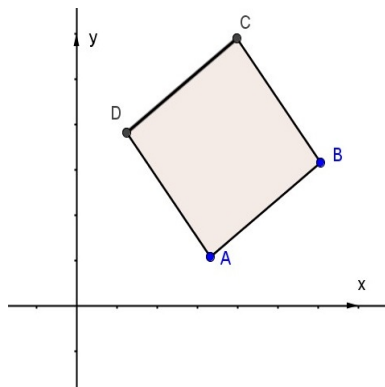
(A) $\bar{\omega}$

(B) ω^2

(C) $\frac{1}{\omega}$

(D) ω^3

10. The Argand plane shows the square ABCD in the first quadrant. The point A represents the complex number z and the point C represents the complex number ω . 1



Which of the following represents the point D?

(A) $\frac{z + \omega}{2} + i\frac{z - \omega}{2}$

(C) $\frac{z + \omega}{2} - i\frac{z - \omega}{2}$

(B) $\frac{z - \omega}{2} + i\frac{z + \omega}{2}$

(D) $\frac{z - \omega}{2} - i\frac{z + \omega}{2}$

Examination continues overleaf...

Section II

90 marks

Attempt Questions 11 to 16

Allow approximately 2 hours and 45 minutes for this section.

Write your answers in the writing booklets supplied. Additional writing booklets are available. Your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 Marks)

Commence a NEW page.

Marks

(a) Evaluate $\int \frac{dx}{\sqrt{3+2x-x^2}}$. **2**

(b) Evaluate $\int_{-1}^0 \frac{2x^3 - 4x + 1}{x - 1} dx$. **3**

(c) By using $t = \tan \frac{x}{2}$, find $\int \frac{dx}{2 + \sin x + \cos x}$. **3**

(d) By using integration by parts, find $\int e^{-x} \cos 2x dx$. **3**

(e) Consider

$$\frac{36}{(x+4)^2(2-x)} = \frac{a}{x+4} + \frac{b}{(x+4)^2} + \frac{c}{2-x}$$

i. Find a , b and c . **2**

ii. Hence or otherwise, evaluate $\int \frac{36}{(x+4)^2(2-x)} dx$. **2**

End of Question 11

Question 12 (15 Marks)	Commence a NEW page.	Marks
(a) Let $z = 1 + i\sqrt{3}$		
i. Find the value of $ z $		1
ii. Express $\frac{\bar{z}}{z}$ in modulus-argument form.		2
(b) Sketch the locus of z if $\frac{z+3i}{z-3i}$ is purely imaginary.		2
(c) $1 - i$ is a root of the quadratic equation $z^2 + \omega z - i = 0$. Find the complex number ω in the form $a + ib$.		2
(d) For the ellipse $\frac{x^2}{9} + \frac{y^2}{16} = 1$, find:		
i. The eccentricity.		1
ii. The coordinates of the foci S and S' and the equations of its directrices.		2
iii. Sketch the ellipse showing all the above features.		2
(e) The polynomial $x^4 - 3x^3 - 2x^2 + 2x + 1 = 0$ has roots α , β and γ . Find an equation with roots $\alpha^2 - 1$, $\beta^2 - 1$ and $\gamma^2 - 1$.		3

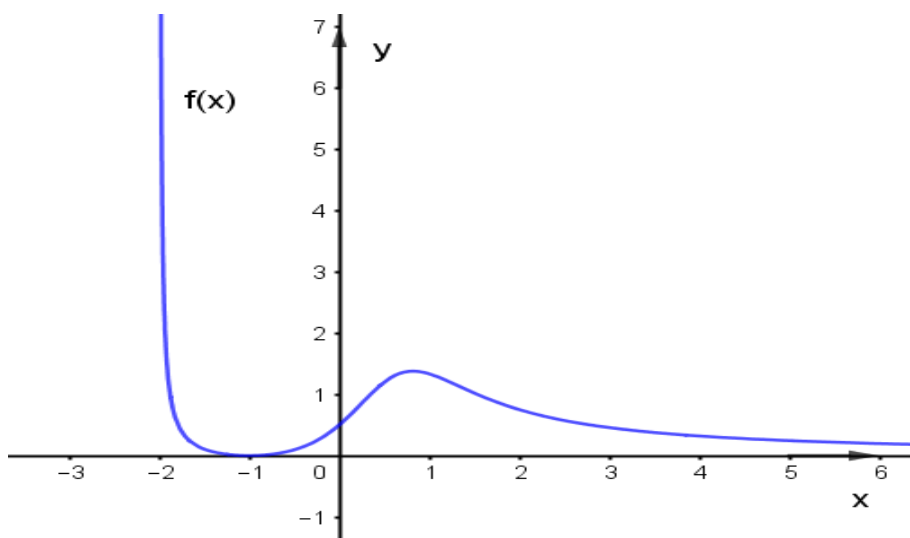
End of Question 12

Question 13 (15 Marks)

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Marks

- (a) The diagram shows the graph of the function $y = f(x)$.

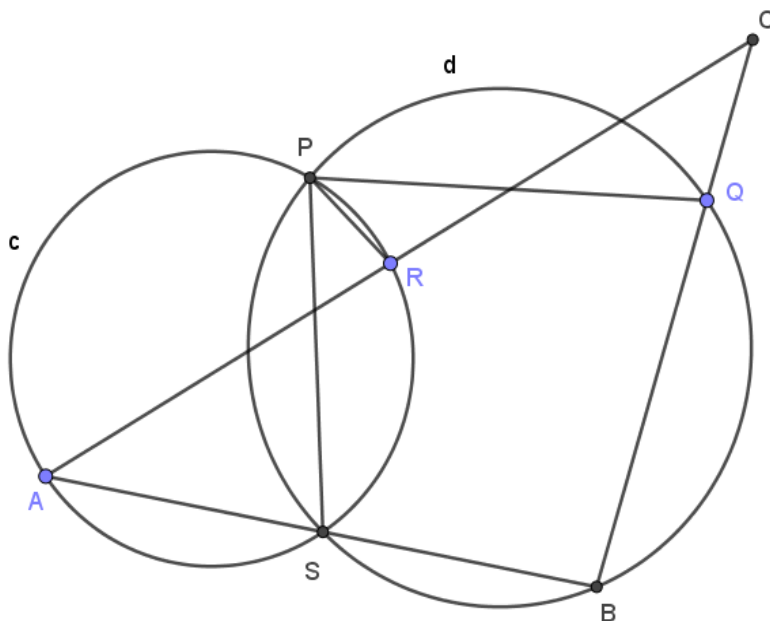


Draw separate one-third page sketches of graphs of the following:

- i. $y = \sqrt{f(x)}$. **2**
 - ii. $y = \frac{1}{f(x)}$. **2**
 - iii. $y = xf(x)$. **2**
- (b) A solid is formed by rotating the area enclosed by the curve $x^2 + y^2 = 9$ through one complete revolution about the line $x = 7$.
- i. By taking slices perpendicular to the axis of rotation, show that the volume of the solid is $V = 28\pi \int_{-3}^3 \sqrt{9 - y^2} dy$. **3**
 - ii. Find the exact volume of the solid. **2**

Question 13 continues on the next page...

- (c) Two circles c and d meet at P and S . Points A and R lie on c and points B and Q lie on d . AB passes through S and AR produced meets BQ produced at C , as shown in the diagram.



- i. **Copy the diagram to your booklet.**
- ii. Prove that $\angle PRA = \angle PQB$. **2**
- iii. Prove that the points P , R , Q and C are concyclic. **2**

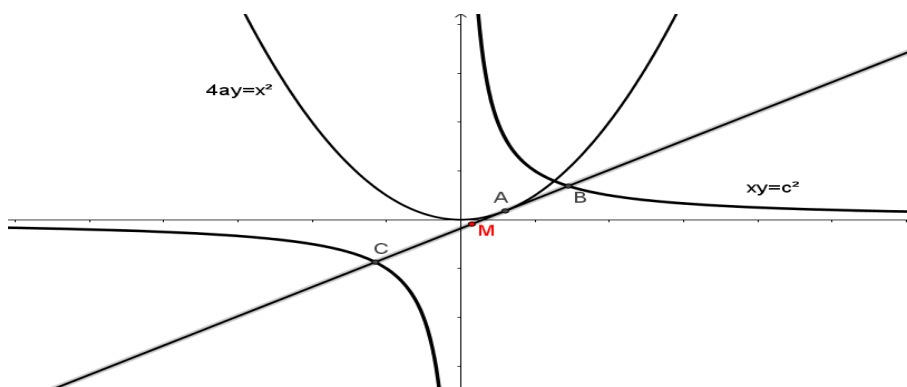
End of Question 13

Question 14 (15 Marks)

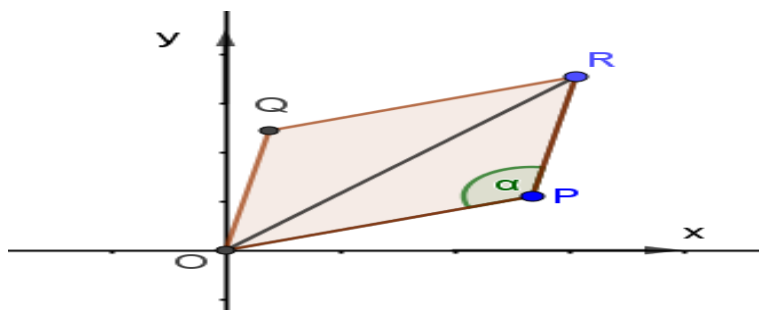
Commence a NEW page.

Marks

- (a) Consider the hyperbola $xy = c^2$ and the parabola $4ay = x^2$. Let $A(2at, at^2)$ lie on the parabola.



- i. Derive the equation of the tangent at A . 2
 - ii. The tangent in (i) cuts the hyperbola at B and C . Without finding the coordinates of B and C , find the coordinates of M the midpoint of BC . 3
 - iii. Hence, find the equation of the locus of M as A moves on the parabola, stating all restrictions. 2
- (b) The diagram shows the parallelogram OQRP in the Argand plane with the point P represented by the complex number z and Q represented by the complex number ω and $\alpha = \angle OPR$.



- i. Show that $\arg\left(\frac{\omega}{z}\right) = \pi - \alpha$ 1
- ii. Use the cosine rule to show that 2

$$|z + \omega|^2 = |z|^2 + |\omega|^2 + 2|z||\omega|\cos\left(\arg\left(\frac{\omega}{z}\right)\right)$$

- iii. Hence or otherwise show that 2

$$\cos\left(\arg\left(\frac{\omega}{z}\right)\right) = \frac{|z + \omega|^2 - |\omega - z|^2}{4|z||\omega|}$$

- iv. If $\alpha = \frac{7\pi}{12}$, $|PR| = 2$ and $z = 3\left(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right)$, find in Cartesian form the complex number $\omega + z$. 3

End of Question 14

Question 15 (15 Marks)

Commence a NEW page.

Marks

(a) Given that $z = \cos \theta + i \sin \theta$ and $z^n - \frac{1}{z^n} = 2i \sin n\theta$.

i. By considering $\left(z - \frac{1}{z}\right)^5$, show that **2**

$$\sin^5 \theta = \frac{1}{16} (\sin 5\theta - 5 \sin 3\theta + 10 \sin \theta).$$

ii. Solve the following equation for $0 \leq \theta \leq 2\pi$ **3**

$$\sin 5\theta - 5 \sin 3\theta + 6 \sin \theta = 0.$$

(b) If $I_n = \int_0^{\frac{\pi}{4}} \sec^n x dx$

i. Show that $I_n = \frac{(\sqrt{2})^{n-2}}{n-1} + \frac{n-2}{n-1} I_{n-2}$, for $n \geq 2$. **3**

ii. Hence or otherwise, evaluate $\int_0^1 (1+x^2)^{\frac{5}{2}} dx$. **3**

(c) Let a and b be real numbers. Consider the cubic equation

$$x^3 - 2bx^2 - a^2x + b^2 = 0$$

i. Show that if $x = -1$ is a solution, then $1 - \sqrt{2} \leq b \leq 1 + \sqrt{2}$. **2**

ii. Show that there is no value of b for which $x = -1$ is a repeated root. **2**

End of Question 15

Question 16 (15 Marks)

Commence a NEW page.

Marks(a) Suppose that x is a positive real number.

i. Find the sum of the geometric series

1

$$1 - t^3 + t^6 - t^9 + \dots + t^{6n}$$

ii. Hence, show that

1

$$\frac{1}{1+t^3} < 1 - t^3 + t^6 - t^9 + \dots + t^{6n}, \quad \text{for } 0 < t < x.$$

iii. Find the sum of the geometric series

1

$$1 - t^3 + t^6 - t^9 + \dots + t^{6n} - t^{6n+3}$$

iv. Hence, show that

1

$$1 - t^3 + t^6 - t^9 + \dots + t^{6n} < \frac{1}{1+t^3} + t^{6n+3}, \quad \text{for } 0 < t < x.$$

v. Multiply the inequalities of part (ii) and (iv) by a suitable factor, then integrating from $t = 0$ to $t = x$, show that**3**

$$\frac{1}{3} \ln(1+x^3) < \frac{x^3}{3} - \frac{x^6}{6} + \dots + \frac{x^{6n+3}}{6n+3} < \frac{1}{3} \ln(1+x^3) + \frac{x^{6(n+1)}}{6(n+1)}.$$

vi. By taking limit as $n \rightarrow \infty$, show that for $0 \leq x \leq 1$ **1**

$$\ln(1+x^3) = x^3 - \frac{x^6}{2} + \frac{x^9}{3} - \frac{x^{12}}{4} + \dots$$

vii. Use suitable substitution to prove that

1

$$\ln 2 = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots$$

viii. Hence or otherwise prove that

2

$$\ln 4 = 1 + \frac{1}{2.3} + \frac{1}{3.5} + \frac{1}{4.7} + \dots$$

(b) Given that a, b and c are all positives such that $a+b+c = 1$ and $a+b+c \geq 3^3\sqrt{abc}$.i. If x, y and z are all positives, show that**1**

$$\frac{1}{xy} + x + y \geq 3$$

ii. Hence or otherwise, prove that

3

$$\frac{1}{a(a+1)} + \frac{1}{b(b+1)} + \frac{1}{c(c+1)} \geq 4$$

End of paper.

Answer sheet for Section I

Mark answers to Section I by fully blackening the correct circle, e.g “●”

STUDENT NUMBER:

Class (please ✓)

☐ 12M4A – Miss Lee

☐ 12M4B – Dr Jomaa

☐ 12M4C – Mr Lin

- | | | | | |
|------|------------------------------------|------------------------------------|------------------------------------|------------------------------------|
| 1 – | ☐ A | ☒ B | ☐ C | ☐ D |
| 2 – | ☒ A | ☐ B | ☐ C | ☐ D |
| 3 – | ☐ A | ☐ B | ☐ C | ☒ D |
| 4 – | ☐ A | ☐ B | ☒ C | ☐ D |
| 5 – | ☐ A | ☐ B | ☐ C | ☒ D |
| 6 – | ☐ A | ☒ B | ☐ C | ☐ D |
| 7 – | ☐ A | ☒ B | ☐ C | ☐ D |
| 8 – | ☐ A | ☐ B | ☐ C | ☒ D |
| 9 – | ☐ A | ☒ B | ☐ C | ☐ D |
| 10 – | ☐ A | ☐ B | ☒ C | ☐ D |

MC:

1. $z = \sqrt{3} + i$

$$\frac{i}{z} = \frac{\bar{i}}{\bar{z}} = \frac{\bar{i}z}{|z|^2} = \frac{-iz}{|z|^2} = \frac{-i(\sqrt{3}+i)}{4} = \frac{1-i\sqrt{3}}{4}$$

(B)

2. $ae = 5$ and $\frac{a}{e} = 10$

$$10e^2 = 5 \therefore e^2 = \frac{1}{2} \therefore e = \frac{1}{\sqrt{2}}$$

(A)

3. $\alpha^3 + \beta^3 + \gamma^3 = \frac{5(\alpha + \beta + \gamma) - 3 \times 6}{3} = -6$

(D)

4. $\frac{2}{x+x^3} = \frac{A}{x} + \frac{Bx+C}{1+x^2} = \frac{A(1+x^2) + (Bx+C)x}{x(1+x^2)} = \frac{Ax^2 + A + Bx^2 + Cx}{x(1+x^2)}$

$$A+B=0, C=0, A=2 \therefore B=-2$$

$$\frac{2}{x+x^3} = \frac{2}{x} - \frac{2x}{1+x^2}$$

$$\int \frac{2 dx}{x+x^3} = 2 \ln|x| - \ln|1+x^2| + C = \ln \left| \frac{x^2}{1+x^2} \right| + C$$

(C)

5. $\int_0^{\pi/3} \sin^4 x (1 - \sin^2 x) \cos x dx = \int_0^{\pi/3} (\sin^4 x - \sin^6 x) \cos x dx$

Let $u = \sin x$ $du = \cos x dx$

$$\left. \begin{array}{l} u = 0 \quad (x=0) \\ u = \frac{\sqrt{3}}{2} \quad (x=\pi/3) \end{array} \right\} = \int_0^{\sqrt{3}/2} (u^4 - u^6) du$$

(D)

(6)

(B)

(7) $3y^2 \frac{dy}{dx} = 3x^2 \frac{dy}{dx} + 6xy - 6x^2$

$$3(y^2 - x^2) \frac{dy}{dx} = 6x(y-x)$$

$$\frac{dy}{dx} = \frac{2x}{y+x}$$

(B)

$$8. \quad p^3 x - py + c - cp^4 = 0$$

$$p^3 q - \frac{p^2}{q} + c - cp^4 = 0$$

$$cp^3 q^2 - cp + cq - cp^4 q = 0$$

$$p^3 q^2 - p^4 q = p - q$$

$$p^3 q(q - p) = p - q \quad \therefore$$

$$\boxed{p^3 q = -1}$$

(D)

9.

(B)

10.

$$\vec{AD} = i \vec{AB} \\ = i \vec{DC}$$

$$z_D - z = i(w - z_D)$$

$$z_D(1+i) = z + iw$$

$$z_D = \frac{z}{1+i} + \frac{iw}{1+i}$$

$$= \frac{z(1-i)}{2} + \frac{i(1-i)w}{2}$$

$$= \frac{z}{2} - i\frac{z}{2} + \frac{i}{2}w + \frac{w}{2}$$

$$= \frac{z+iw}{2} + i\frac{(z-w)}{2}$$

(C)

Question 11:

$$(a) \int \frac{dx}{\sqrt{3+2x-x^2}} = \int \frac{dx}{\sqrt{4-(x-1)^2}} = \boxed{\sin^{-1}\left(\frac{x-1}{2}\right) + C}$$

$$\begin{aligned}(b) \int_{-1}^0 \frac{2x^3-4x+1}{x-1} dx &= \int_{-1}^0 \frac{2x^3-2x-2x+2-1}{x-1} dx \\&= 2 \int_{-1}^0 \frac{x(x^2-1)}{x-1} dx - 2 \int_{-1}^0 \frac{x-1}{x-1} dx - \int_{-1}^0 \frac{dx}{x-1} \quad \checkmark \\&= 2 \int_{-1}^0 x(x+1) dx - 2 \int_{-1}^0 dx - \int_{-1}^0 \frac{dx}{x-1} \\&= 2 \left[\frac{x^3}{3} + \frac{x^2}{2} \right]_{-1}^0 - 2[x]_{-1}^0 - \left| \ln|x-1| \right|_{-1}^0 \quad \checkmark \\&= 2\left(0 - \left(-\frac{1}{3} + \frac{1}{2}\right)\right) - 2(0 - (-1)) - (\ln 1 - \ln 2) \\&= \boxed{-\frac{1}{3} - 2 + \ln 2} \quad \checkmark\end{aligned}$$

$$\begin{aligned}(c) \quad t &= \tan \frac{x}{2}, \quad dt = \frac{1}{2}(1+t^2) dx \therefore dx = \frac{2dt}{1+t^2} \\2 + \sin x + \cos x &= 2 + \frac{2t}{1+t^2} + \frac{1-t^2}{1+t^2} = \frac{2+2t^2+2t+1-t^2}{1+t^2} = \frac{t^2+2t+3}{1+t^2} \\ \frac{1}{2+\sin x+\cos x} &= \frac{1+t^2}{t^2+2t+3} \quad \checkmark \\\int \frac{1}{2+\sin x+\cos x} dx &= \int \frac{1+t^2}{t^2+2t+3} \times \frac{2dt}{1+t^2} = 2 \int \frac{dt}{(t+1)^2+2} \quad \checkmark \\&= 2 \times \frac{1}{\sqrt{2}} \tan^{-1}\left(\frac{t+1}{\sqrt{2}}\right) + C \\&= \boxed{\sqrt{2} \tan^{-1}\left(\frac{1+\tan \frac{x}{2}}{\sqrt{2}}\right) + C} \quad \checkmark\end{aligned}$$

$$(d) \quad I = \int e^{-x} \cos x \, dx$$

$$\text{let } u = \cos x \quad \therefore \, du = -\sin x \, dx$$

$$e^{-x} dx = dv \quad \therefore \, v = -e^{-x}$$

$$I = uv - \int v \, du$$

$$= -e^{-x} \cos x - 2 \int e^{-x} \sin x \, dx \quad \checkmark$$

$$\text{let } u = \sin x \quad \therefore \, du = \cos x \, dx$$

$$e^{-x} dx = dv \quad \therefore \, v = -e^{-x}$$

$$I = -e^{-x} \cos x - 2 \left[-e^{-x} \sin x + 2 \int e^{-x} \cos x \, dx \right] \quad \checkmark$$

$$= -e^{-x} \cos x + 2e^{-x} \sin x - 4I$$

$$5I = e^{-x} (2 \sin 2x - \cos x)$$

$$\boxed{I = \frac{e^{-x}}{5} (2 \sin 2x - \cos x)} \quad \checkmark$$

$$(e) \quad \frac{36}{(x+4)^2(2-x)} = \frac{a}{x+4} + \frac{b}{(x+4)^2} + \frac{c}{2-x}$$

(i) multiply by $(x+4)^2$ then set $x = -4$

$$\frac{36}{6} = b \quad \therefore \quad \boxed{b = 6}$$

multiply by $2-x$ then set $x = 2$ ✓

$$\frac{36}{36} = c \quad \therefore \quad \boxed{c = 1}$$

$$\text{Set } x = 0 \quad \therefore \quad \frac{36}{16 \times 2} = \frac{a}{4} + \frac{b}{16} + \frac{c}{2}$$

$$3 = a + \frac{b}{4} + 2c$$

$$a = 1 - \frac{3}{2} = \boxed{-\frac{1}{2}} \quad \checkmark$$

$$\begin{aligned} \text{(ii)} \quad \int \frac{36}{(x+4)^2(2-x)} dx &= \frac{-1}{2} \int \frac{dx}{x+4} + 6 \int \frac{dx}{(x+4)^2} + \int \frac{dx}{2-x} \quad \checkmark \\ &= \left[\frac{-1}{2} \ln|x+4| - \frac{6}{x+4} - \ln|2-x| + C \right] \quad \checkmark \end{aligned}$$

Question 12

(a) $z = 1 + i\sqrt{3}$

(i) $|z| = \sqrt{1^2 + (\sqrt{3})^2} = \sqrt{4} = 2$

(ii) $\frac{\bar{z}}{z} = \frac{\bar{z}}{z} \times \frac{\bar{z}}{\bar{z}} = \frac{(\bar{z})^2}{|z|^2} = \frac{(1 - i\sqrt{3})^2}{4} = \left(\frac{1}{2} - i\frac{\sqrt{3}}{2}\right)^2 = \left[\cos\left(-\frac{\pi}{3}\right)\right]^2$

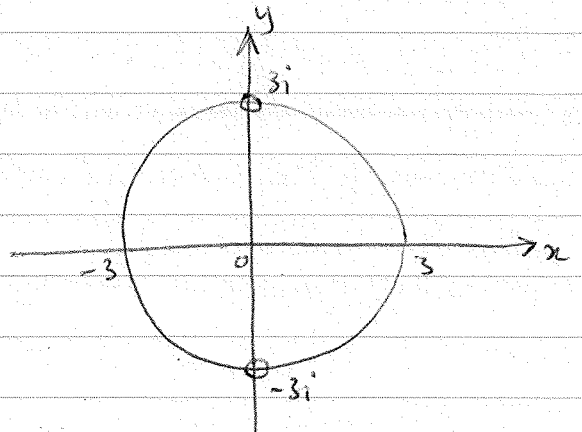
$\frac{\bar{z}}{z} = \cos\left(-\frac{2\pi}{3}\right) = \cos\left(-\frac{2\pi}{3}\right) + i\sin\left(-\frac{2\pi}{3}\right)$

(b) $\frac{z+3i}{z-3i}$ is purely imaginary

$\therefore \arg\left(\frac{z+3i}{z-3i}\right) = \pm \pi/2$

so the locus of z is the circle

$x^2 + y^2 = 9$ excluding $(0, 3)$ and $(0, -3)$.



Algebraically, $\frac{z+3i}{z-3i} = \frac{x+i(y+3)}{x+i(y-3)}$

Let $z = x+iy$ $\frac{z+3i}{z-3i} = \frac{x+i(y+3)}{x+i(y-3)}$

$= \frac{(x+i(y+3))(x-i(y-3))}{x^2 + (y-3)^2}$

$\frac{z+3i}{z-3i} = \frac{x^2 + (y+3)(y-3) + i(xy + 3x - yx + 3x)}{x^2 + (y-3)^2}$

$= \frac{x^2 + y^2 - 9 + 6xi}{x^2 + (y-3)^2}$

$\frac{z+3i}{z-3i}$ is purely imaginary $\therefore \operatorname{Re}\left(\frac{z+3i}{z-3i}\right) = 0 \therefore x^2 + y^2 - 9 = 0$

$\therefore \boxed{x^2 + y^2 = 9}$ excluding $z = \pm 3i$

(c) $(1-i)^n + w(1-i) - i = 0$

$-2i + w(1-i) - i = 0 \therefore w = \frac{3i}{1-i} = \frac{3i(1+i)}{2}$

$\boxed{w = -\frac{3}{2} + \frac{3}{2}i}$

d)

$$a = 3, b = 4.$$

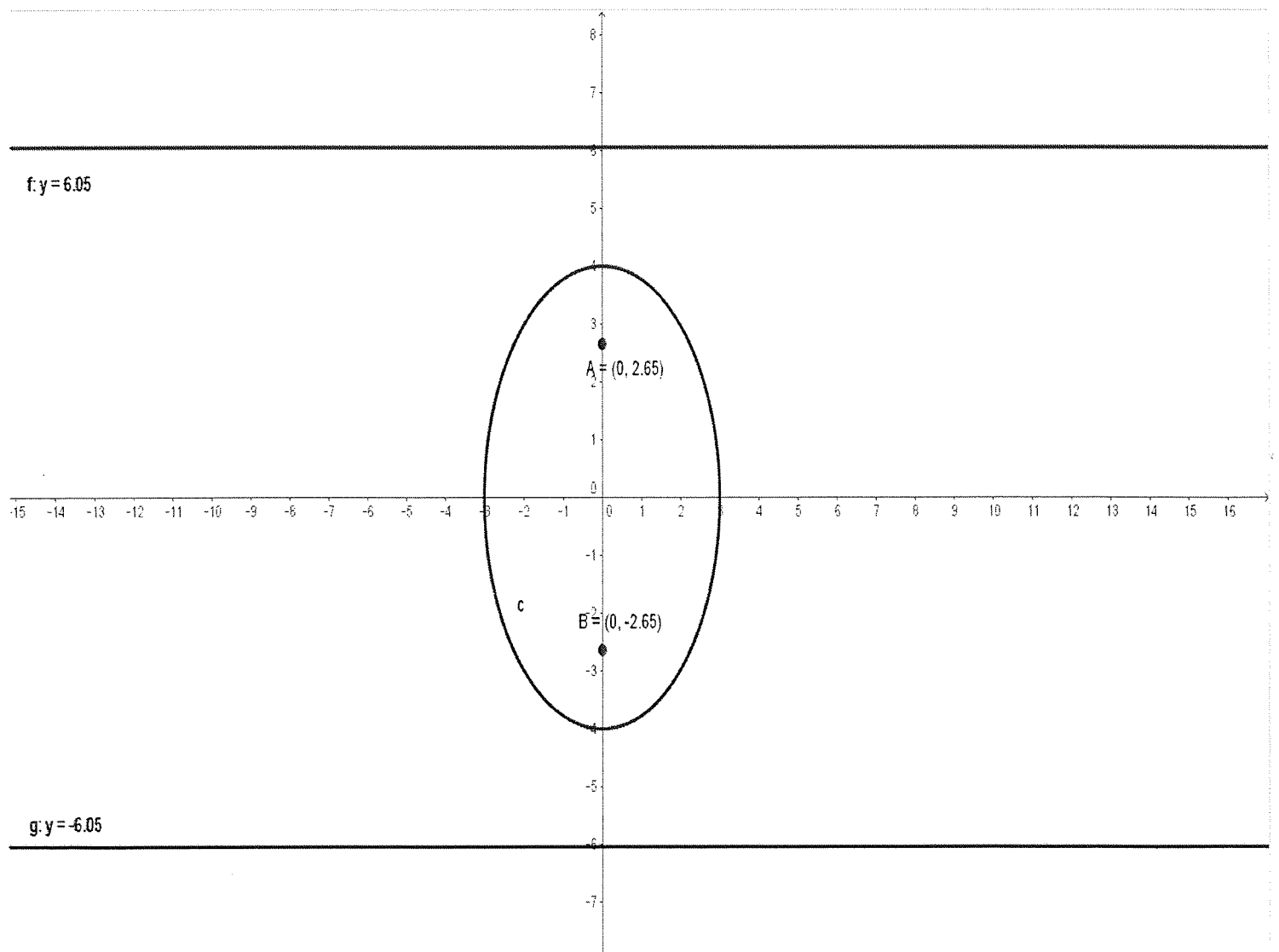
i. $a^2 = b^2 * (1 - e^2)$

$$e = \sqrt{1 - \frac{a^2}{b^2}} = \frac{\sqrt{7}}{4}$$

ii. Foci: $S(0, \pm\sqrt{7})$

Directrices: $y = \pm \frac{16\sqrt{7}}{7}$

iii.



② Let $x^2 - 1 = y \therefore x^2 = y + 1$
 $x^4 = (y + 1)^2$
 $x^3 = (y + 1)\sqrt{y + 1}$
 $x = \sqrt{y + 1}$

x is a root of $x^4 - 3x^3 - 2x^2 + 2x + 1 = 0$

$$\therefore (y + 1)^2 - 3(y + 1)\sqrt{y + 1} - 2(y + 1) + 2\sqrt{y + 1} + 1 = 0$$

$$(y + 1)^2 - 2(y + 1) + 1 = 3(y + 1)\sqrt{y + 1} - 2\sqrt{y + 1}$$

$$(y + 1 - 1)^2 = (3y + 3 - 2)\sqrt{y + 1}$$

$$y^2 = (3y + 1)\sqrt{y + 1}$$

$$(y^2)^2 = (3y + 1)^2 (\sqrt{y + 1})^2$$

$$y^4 = (9y^2 + 6y + 1)(y + 1)$$

$$= 9y^3 + 9y^2 + 6y^2 + 6y + y + 1$$

$$= 9y^3 + 15y^2 + 7y + 1$$

$$\therefore y^4 - 9y^3 - 15y^2 - 7y - 1 = 0$$

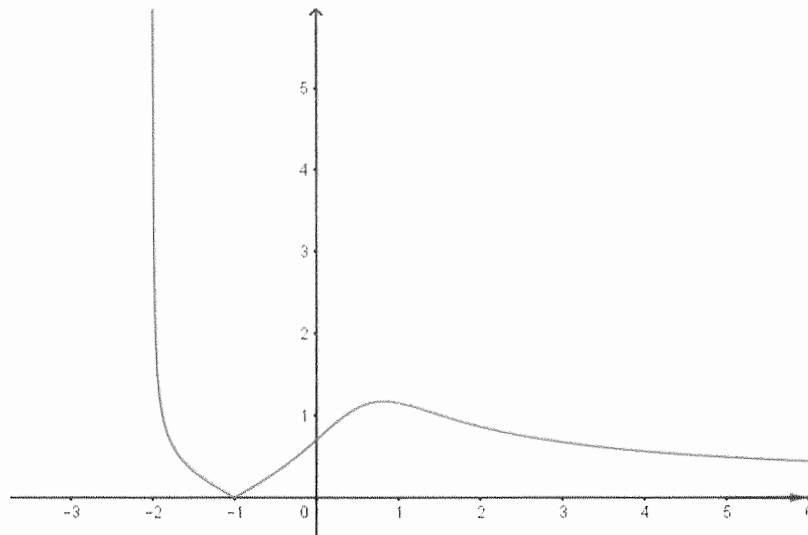
$\therefore$ The equation which has roots $x^2 - 1, \beta^2 - 1, \gamma^2 - 1, \delta^2 - 1$ is

$$\boxed{x^4 - 9x^3 - 15x^2 - 7x - 1 = 0}$$

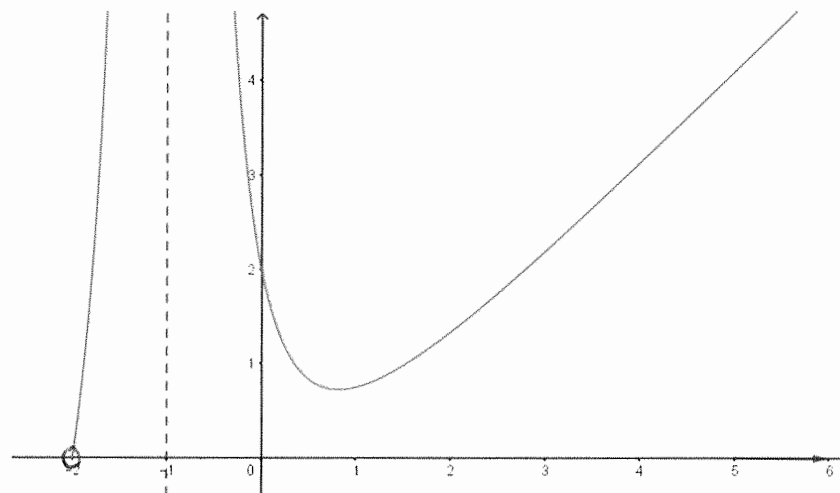
Question 13

(a)

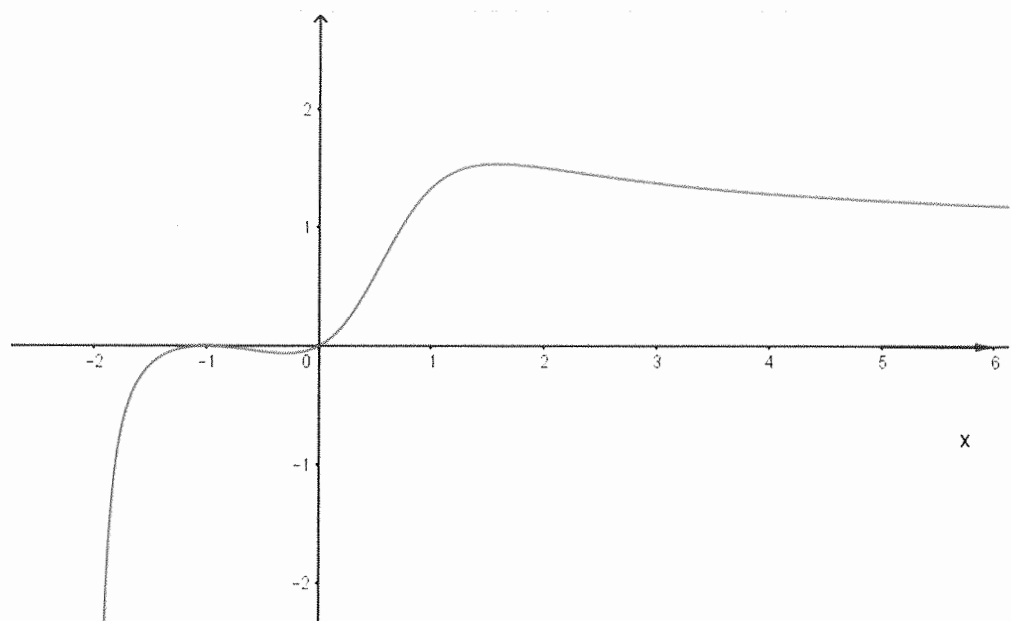
(i) $y = \sqrt{f(x)}$



(ii) $y = \frac{1}{f(x)}$

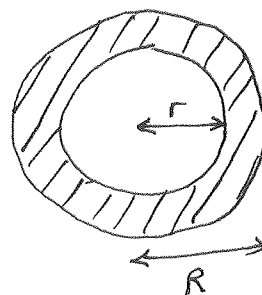
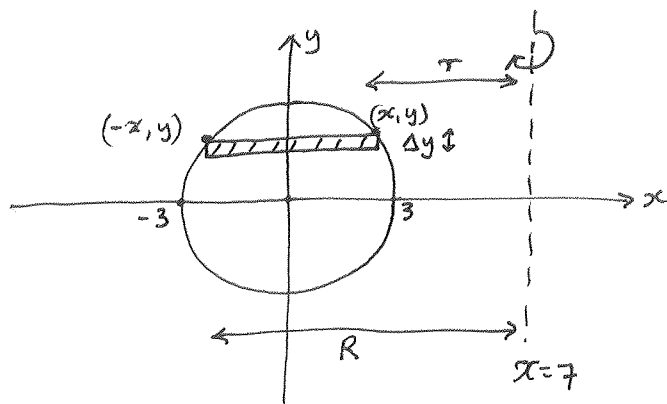


(iii) $y = xf(x)$



13(b)

(i)



$$r = 7 - x = 7 - \sqrt{9 - y^2}$$

$$R = 7 + x = 7 + \sqrt{9 - y^2}$$

(R and r are the outer & inner radii when the hatched strip is rotated about line $x = 7$.)

Area of annular cross section is

$$\begin{aligned} \Delta A &= \pi R^2 - \pi r^2 = \pi (R+r)(R-r) \\ &= \pi (14) (2\sqrt{9-y^2}) \end{aligned}$$

$$\therefore \Delta A = 28\pi \sqrt{9-y^2}$$

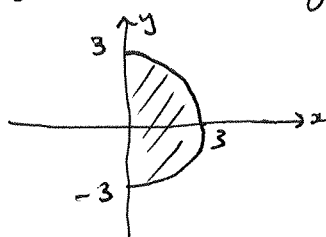
Let slice thickness be Δy . Then

$$\text{Volume element is } \Delta V = 28\pi \sqrt{9-y^2} \cdot \Delta y$$

$$\therefore V = \lim_{\Delta y \rightarrow 0} \sum_{-3}^3 28\pi \sqrt{9-y^2} \cdot \Delta y = 28\pi \int_{-3}^3 \sqrt{9-y^2} dy$$

(ii)

$\int_{-3}^3 \sqrt{9-y^2} dy$ is the area of a semicircle:



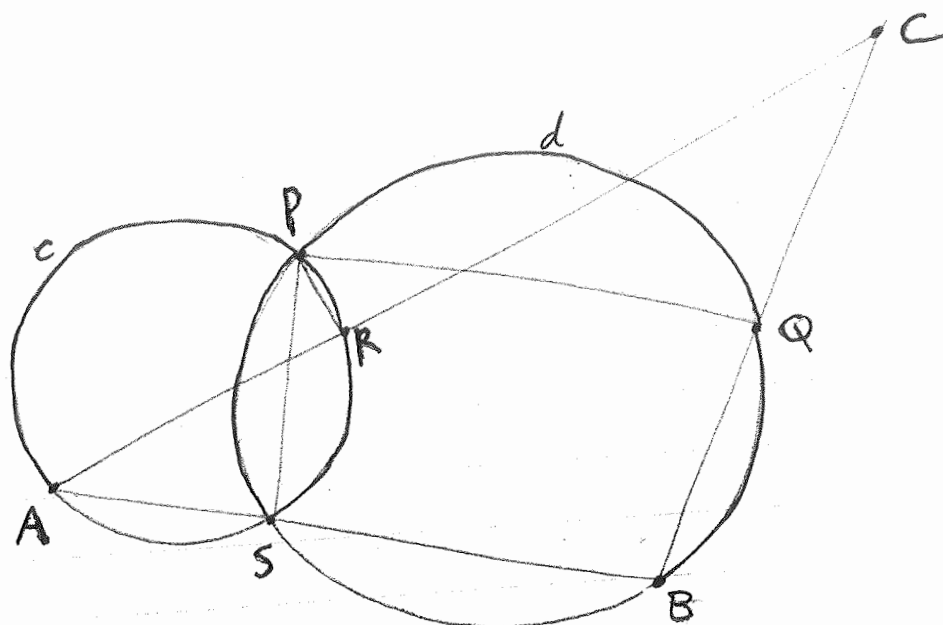
$$\therefore \text{it equals } \frac{1}{2} \times \pi \times 3^2$$

$$\therefore V = 28\pi \times \frac{1}{2} \times \pi \times 3^2$$

$$\therefore V = 126\pi^2 \text{ units}^3$$

[ALT: the integral may be done from scratch, using a substitution $y = 3 \sin \theta$.]

(i)



✓✓

$$\therefore 180 - \angle PRA = 180 - \angle PQA$$

$\therefore$ PRQC is cyclic (interval PC subtends equal angles at two points on the same side of it).

✓✓

Question 14: (a) $xy = c^2$ and $4ay = x^2$, $A(2at, at^2)$

(i) $y' = \frac{2x}{4a} = \frac{x}{2a}$ at $x = 2at$ $y' = t$ ✓

$$y - at^2 = t(x - 2at)$$

$$y = tx - 2at^2 + at^2$$

$$\boxed{y = tx - at^2} \quad (1) \quad \checkmark$$

(ii) Tangent and hyperbola meet at B and C

Multiply (1) by x, we obtain

$$xy = tx^2 - at^2x, \text{ but } xy = c^2$$

[alt: solve simultaneously]

$$\therefore \boxed{tx^2 - at^2x - c^2 = 0} \quad (2) \quad \checkmark$$

The solution of equation (2) are the x -coordinates of B and C. So the sum of roots equal at and the x -coordinate of M the midpoint of BC ✓
is $\left(\frac{1}{2}at\right)$

M lies on the tangent $\therefore$ M Satisfy equation (1)

$$y = t\left(\frac{1}{2}at\right) - at^2 = \frac{1}{2}at^2 - at^2 = -\frac{1}{2}at^2$$

So $\boxed{M\left(\frac{1}{2}at, -\frac{1}{2}at^2\right)}$ ✓

iii) $x = \frac{1}{2}at \therefore t = \frac{2x}{a}$

$$y = -\frac{1}{2}at^2 = -\frac{1}{2}a\left(\frac{2x}{a}\right)^2 = -\frac{1}{2}a \times \frac{4x^2}{a^2} = \boxed{\frac{-2x^2}{a}} \quad \checkmark$$

In Equation (2) the roots equal $\frac{at^2 \pm \sqrt{a^2t^4 + 4c^2}}{2t}$

Since t takes all values $\therefore a^2t^4 + 4c^2t$ can not be negative

$\therefore a^2\left(\frac{2x}{a}\right)^4 + 4c^2\left(\frac{2x}{a}\right)$ can not be negative

$$\frac{16x^4}{a^2} + \frac{8c^2x}{a} = \frac{8}{a^2}(2x^4 + ac^2x) = \frac{8}{a^2}x(2x^3 + ac^2) \geq 0$$

$$\boxed{x < \left(\frac{-ac^2}{2}\right)^{1/3} \text{ or } x \geq 0} \quad \checkmark$$

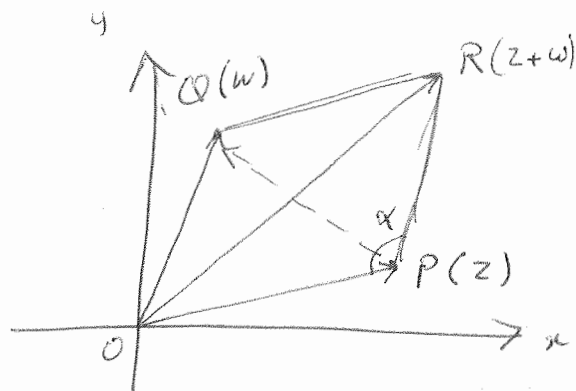
(b)

$$(i) \angle POQ = \arg(w) - \arg(z) \\ = \arg\left(\frac{w}{z}\right)$$

OPRQ is a parallelogram

$\therefore \angle POQ$ and $\angle OPR$ are supplementary

$$\therefore \arg\left(\frac{w}{z}\right) + \alpha = \pi \\ \text{and } \arg\left(\frac{w}{z}\right) = \pi - \alpha$$



(ii) $\vec{OR}$ represents $z+w$

$$|z+w|^2 = |z|^2 + |w|^2 - 2|z||w|\cos\alpha \\ = |z|^2 + |w|^2 - 2|z||w|\cos(\pi - \arg(\frac{w}{z})) \\ = |z|^2 + |w|^2 + 2|z||w|\cos(\arg(\frac{w}{z})) \quad (1)$$

(iii) in $\triangle OPQ$, $\vec{PQ}$ represents $w-z$

$$|w-z|^2 = |w|^2 + |z|^2 - 2|w||z|\cos(\arg(\frac{w}{z})) \quad (2)$$

$$(1) - (2) \therefore |z+w|^2 - |w-z|^2 = 4|z||w|\cos(\arg(\frac{w}{z}))$$

$$\therefore \cos(\arg(\frac{w}{z})) = \frac{|z+w|^2 - |w-z|^2}{4|z||w|}$$

$$(iv) \alpha = \frac{7\pi}{12}, |PR| = 2, z = 3\left(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right), \arg(z) = \frac{\pi}{4}$$

$$\arg\left(\frac{w}{z}\right) = \pi - \alpha = \pi - \frac{7\pi}{12} = \frac{5\pi}{12}$$

$$\arg(w) = \arg(z) + \frac{5\pi}{12} = \frac{\pi}{4} + \frac{5\pi}{12} = \frac{2\pi}{3}$$

$$\text{and } |w| = 2 \therefore w = 2\left(\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3}\right)$$

$$w+z = 2\cos\frac{2\pi}{3} + 3\cos\frac{\pi}{4} \\ = 2\left(-\frac{1}{2} + i\frac{\sqrt{3}}{2}\right) + 3\left(\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}i\right) \\ = -1 + \frac{3}{\sqrt{2}} + i\left(\sqrt{3} + \frac{3}{\sqrt{2}}\right)$$

Question 15: $z = \cos \theta$, $z^n - \frac{1}{z^n} = 2i \sin n\theta$

(i) $\left(z - \frac{1}{z}\right)^5 = z^5 - 5z^3 + 10z - \frac{1}{10}z + 5\frac{1}{z} - \frac{1}{z^5}$

$|z| = 1$, $\frac{1}{z} = \bar{z}$, $z - \bar{z} = 2i \sin \theta$

$\left(z - \frac{1}{z}\right)^5 = (2i \sin \theta)^5 = z^5 - \frac{1}{z^5} - 5\left(z^3 - \frac{1}{z^3}\right) + 10\left(z - \frac{1}{z}\right)$

$2^5 i \sin^5 \theta = 2i \sin 5\theta - 5(2i \sin 3\theta) + 10(2i \sin \theta)$

$\sin^5 \theta = \frac{1}{16}(\sin 5\theta - 5 \sin 3\theta + 10 \sin \theta)$ (*)

(ii) $\sin 5\theta - 5 \sin \theta + 6 \sin \theta = 0$

but $\sin 5\theta - 5 \sin \theta + 6 \sin \theta = 16 \sin^5 \theta - 4 \sin \theta$ from (*)

$\therefore 4 \sin \theta (4 \sin^4 \theta - 1) = 0$

$\therefore 4 \sin \theta (2 \sin^2 \theta - 1)(2 \sin^2 \theta + 1) = 0$

$\therefore \sin \theta = 0$ or $\sin \theta = \pm \frac{1}{\sqrt{2}}$

$\theta = 0, \pi, 2\pi, \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}$ and $\frac{7\pi}{4}$.

(b) $I_n = \int_0^{\pi/4} \sec^n x dx = \int_0^{\pi/4} \sec^{n-2} x \sec^2 x dx$

(i)

let $u = \sec^{n-2} x$ and $\sec^2 x dx = dv$

$du = (n-2) \sec^{n-3} x \times \sec x \tan x dx$ and $v = \tan x$

$I_n = \left[\sec^{n-2} x \tan x \right]_0^{\pi/4} - (n-2) \int_0^{\pi/4} \sec^{n-2} x \tan^2 x dx$

$= (\sqrt{2})^{n-2} - (n-2) \int_0^{\pi/4} \sec^{n-2} x (\sec^2 x - 1) dx$

$= (\sqrt{2})^{n-2} - (n-2) \int_0^{\pi/4} \sec^n x dx + (n-2) \int_0^{\pi/4} \sec^{n-2} x dx$

$= (\sqrt{2})^{n-2} - (n-2) I_n + (n-2) I_{n-2}$

$(n-1) I_n = (\sqrt{2})^{n-2} + (n-2) I_{n-2} \quad \therefore \boxed{I_n = \frac{(\sqrt{2})^{n-2}}{n-1} + \frac{n-2}{n-1} I_{n-2}}$

Q15

(b) (i)

$$I_n = \int_0^{\pi/4} \sec^n x \, dx$$

$$= \int_0^{\pi/4} \sec^2 x \sec^{n-2} x \, dx$$

$$= \int_0^{\pi/4} (\tan^2 x + 1) \sec^{n-2} x \, dx$$

$$= \underbrace{\int_0^{\pi/4} \tan^2 x \sec^{n-2} x \, dx}_{\text{}} + \int_0^{\pi/4} \sec^{n-2} x \, dx$$

$$u = \tan x \quad dv = \tan x \sec^{n-2} x \, dx$$

$$du = \sec^2 x \, dx \quad v = \frac{\sec^{n-2} x}{n-2}$$

$$\therefore I_n = \left[\frac{\tan x \sec^{n-2} x}{n-2} \right]_0^{\pi/4} - \frac{1}{n-2} \int \sec^n x \, dx + I_{n-2}$$

$$= \frac{\sqrt{2}^{n-2}}{n-2} - \frac{1}{n-2} I_n + I_{n-2}$$

$$(n-2) I_n = \sqrt{2}^{n-2} - I_n + (n-2) I_{n-2}$$

$$(n-2+1) I_n = \sqrt{2}^{n-2} + (n-2) I_{n-2}$$

$$I_n = \frac{\sqrt{2}^{n-2}}{n-1} + \frac{n-2}{n-1} I_{n-2}$$

$$(ii) \int_0^1 (1+x^2)^{5/2} dx$$

$$\text{let } x = \tan u \therefore dx = \sec^2 u \, du$$

$$(1+x^2)^{5/2} = (1+\tan^2 u)^{5/2} = (\sec^2 u)^{5/2} = \sec^5 u$$

$$\text{for } x=0 \therefore u=0$$

$$x=1 \therefore u = \pi/4$$

$$\int_0^1 (1+x^2)^{5/2} dx = \int_0^{\pi/4} \sec^5 u \sec^2 u \, du = \int_0^{\pi/4} \sec^7 u \, du = I_7$$

$$\text{Use (i)} \quad I_7 = \frac{(\sqrt{2})^5}{6} + \frac{5}{6} I_5$$

$$I_5 = \frac{(\sqrt{2})^3}{4} + \frac{3}{4} I_3$$

$$I_3 = \frac{(\sqrt{2})}{2} + \frac{1}{2} I_1$$

$$\begin{aligned} I_1 &= \int_0^{\pi/4} \sec u \, du = \int_0^{\pi/4} \frac{\sec u (\sec u + \tan u)}{\sec u + \tan u} \, du \\ &= \left[\ln |\sec u + \tan u| \right]_0^{\pi/4} = \ln(1+\sqrt{2}) \end{aligned}$$

$$I_3 = \frac{\sqrt{2}}{2} + \frac{\ln(1+\sqrt{2})}{2}$$

$$I_5 = \frac{\sqrt{2}}{2} + \frac{3}{4} \left(\frac{\sqrt{2}}{2} + \frac{\ln(1+\sqrt{2})}{2} \right) = \frac{7\sqrt{2}}{8} + \frac{3\ln(1+\sqrt{2})}{8}$$

$$\begin{aligned} I_7 &= \frac{2\sqrt{2}}{3} + \frac{5}{6} \left(\frac{7\sqrt{2}}{8} + \frac{3\ln(1+\sqrt{2})}{8} \right) \\ &= \frac{67\sqrt{2}}{48} + \frac{15\ln(1+\sqrt{2})}{48} \end{aligned}$$

$$(c) \quad x^3 - 2bx^2 - a^2x + b^2 = 0$$

(i) $x = -1$ is a solution $\therefore$

$$(-1)^3 - 2b(-1)^2 - a^2(-1) + b^2 = 0$$

$$-1 - 2b + a^2 + b^2 = 0$$

$$a^2 = 1 + 2b - b^2 = 2 - (b+1)^2 \geq 0 \quad \text{since } a^2 \geq 0$$

$$\therefore -\sqrt{2} \leq b+1 \leq \sqrt{2}$$

$$\therefore 1 - \sqrt{2} \leq b \leq 1 + \sqrt{2}$$

(ii) $x = -1$ is a repeated root

$\therefore x = -1$ is a solution of $x^3 - 2bx^2 - a^2x + b^2 = 0$

Also $x = -1$ is a solution of $3x^2 - 4bx - a^2 = 0$ (the derivative)

$$\therefore -1 - 2b + a^2 + b^2 = 0 \quad \text{from (i)}$$

$$\text{and } 3(-1)^2 - 4b(-1) - a^2 = 0 \therefore 3 + 4b - a^2 = 0$$

$$\text{Now } -1 - 2b + a^2 + b^2 = 0 \quad (1)$$

$$3 + 4b - a^2 = 0 \quad (2)$$

$$(1) + (2) \therefore 2 + 2b + b^2 = 0$$

$$(b+1)^2 + 1 = 0 \quad (3)$$

but b is a real number, hence there is no value of b which satisfy equation (3).

Question 16:

(a) $x > 0$ real number

(i) $1 - t^3 + t^6 - t^9 + \dots + t^{6n}$

$2n+1$ terms of geometric series, their sum equal

$$\frac{1 - (-t^3)^{2n+1}}{1 - (-t^3)} = \frac{1 + (t^3)^{2n+1}}{1 + t^3}$$

(ii) $1 - t^3 + t^6 - t^9 + \dots + t^{6n} = \frac{1 + (t^3)^{2n+1}}{1 + t^3} = \frac{1}{1 + t^3} + \frac{(t^3)^{2n+1}}{1 + t^3} > \frac{1}{1 + t^3}$

$$\therefore \boxed{\frac{1}{1 + t^3} < 1 - t^3 + t^6 - \dots + t^{6n}} \quad (1)$$

(iii) $1 - t^3 + t^6 - t^9 + \dots + t^{6n} - t^{6n+3}$

$2n+2$ terms of geometric series, their sum equal

$$\frac{1 - (-t^3)^{2n+2}}{1 + t^3} = \frac{1 - (t^3)^{2n+2}}{1 + t^3}$$

(iv) $1 - t^3 + t^6 - t^9 + \dots + t^{6n} - t^{6n+3} = \frac{1 - (t^3)^{2n+2}}{1 + t^3} = \frac{1}{1 + t^3} - \frac{(t^3)^{2n+2}}{1 + t^3} < \frac{1}{1 + t^3}$

$$\therefore 1 - t^3 + t^6 - t^9 + \dots + t^{6n} - t^{6n+3} < \frac{1}{1 + t^3}$$

$$\text{ad } \boxed{1 - t^3 + t^6 - t^9 + \dots + t^{6n} < \frac{1}{1 + t^3} + t^{6n+3}} \quad (2)$$

(v) (1) and (2) $\therefore$

$$\frac{1}{1 + t^3} < 1 - t^3 + t^6 - t^9 + \dots + t^{6n} < \frac{1}{1 + t^3} + t^{6n+3} \quad (3)$$

Multiply (3) by t^2 , we obtain

$$\frac{t^2}{1 + t^3} < t^2 - t^5 + t^8 - t^{11} + \dots + t^{6n+2} < \frac{t^2}{1 + t^3} + t^{6n+5} \quad (4)$$

$$\int_0^x \frac{t^2}{1+t^3} dt < \int_0^x (t^2 - t^5 + t^8 - t^{11} + \dots + t^{6n+2}) dt < \int_0^x \frac{t^2}{1+t^3} dt + \int_0^x t^{6n+5} dt$$

$$\left[\frac{1}{3} \ln(1+t^3) \right]_0^x < \left[\frac{t^3}{3} - \frac{t^6}{6} + \frac{t^9}{9} - \frac{t^{12}}{12} + \dots + \frac{t^{6n+3}}{6n+3} \right]_0^x < \left[\frac{1}{3} \ln(1+t^3) \right]_0^x + \left[\frac{t^{6n+6}}{6n+6} \right]_0^x$$

$$\frac{1}{3} \ln(1+x^3) < \frac{x^3}{3} - \frac{x^6}{6} + \frac{x^9}{9} - \frac{x^{12}}{12} + \dots + \frac{x^{6n+3}}{6n+3} < \frac{1}{3} \ln(1+x^3) + \frac{x^{6n+6}}{6n+6}$$

(Vi) As $n \rightarrow \infty$

$$\frac{1}{3} \ln(1+x^3) < \frac{x^3}{3} - \frac{x^6}{6} + \dots < \frac{1}{3} \ln(1+x^3) + 0$$

$$\therefore \ln(1+x^3) = x^3 - \frac{x^6}{2} + \frac{x^9}{3} - \frac{x^{12}}{4} + \dots$$

(Vii) take $x=1$

$$\ln 2 = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots$$

$$(Viii) \ln 2 = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots$$

Combining 1 and $-\frac{1}{2}$, $\frac{1}{3}$ and $-\frac{1}{4}$ and so on, we obtain:

$$\ln 2 = \frac{1}{2} + \frac{1}{3 \cdot 4} + \frac{1}{5 \cdot 6} + \dots$$

$$2 \ln 2 = 1 + \frac{1}{2 \cdot 3} + \frac{1}{3 \cdot 5} + \frac{1}{4 \cdot 7} + \dots$$

$$\ln 4 = 1 + \frac{1}{2 \cdot 3} + \frac{1}{3 \cdot 5} + \frac{1}{4 \cdot 7} + \dots$$

□

$$(b) (i) \frac{1}{xy} + x + y \geq 3 \sqrt[3]{\frac{1}{xy} \times x \times y} = 3$$

$$(ii) \frac{1}{a(a+1)} + \frac{1}{b(b+1)} + \frac{1}{c(c+1)} \geq 4 \quad ?$$

$$\frac{1}{a(a+1)} + a + a+1 \geq 3 \quad (i) \text{ replacing } x \text{ by } a \text{ and } y \text{ by } a+1 \text{ in (i)}$$

Similarly

$$\frac{1}{b(b+1)} + b + b+1 \geq 3 \quad (2)$$

$$\frac{1}{c(c+1)} + c + c+1 \geq 3 \quad (3)$$

$$(1) + (2) + (3) \quad \therefore$$

$$\frac{1}{a(a+1)} + \frac{1}{b(b+1)} + \frac{1}{c(c+1)} + a+b+c + a+b+c + 3 \geq 3+3+3$$

$$\frac{1}{a(a+1)} + \frac{1}{b(b+1)} + \frac{1}{c(c+1)} + 1 + 1 + 3 \geq 9 \quad \text{since } a+b+c=1$$

$$\frac{1}{a(a+1)} + \frac{1}{b(b+1)} + \frac{1}{c(c+1)} \geq 9-5 = 4$$

□